

LSSO: Solved Contextual Operators for Bidirectional Sequence Modeling

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Abstract

Test-time training constructs an input-specific memory by adapting an internal model to contextual associations. For full-context tasks, all positions can participate in this adaptation, but computing the state through repeated updates introduces a tradeoff between update budget and execution cost. We propose LSSO, a bidirectional sequence layer that obtains a shared memory by solving a compact linear system. The system is generated from the complete input and has an exact interpretation as regularized associative-memory error correction with skew-symmetric coupling. Its structure guarantees a unique solution and bounds the forward and adjoint solves. A shared frame maps the state back to tokens, yielding a mixer whose spectral norm is strictly below one and can be computed in compact space for each input. Forward and backward costs grow linearly with sequence length at fixed rank. On eight GenomicBenchmarks tasks with matched encoders and three seeds, LSSO achieves the highest mean accuracy on seven tasks and a macro accuracy of 78.53%, compared with 75.57% for the strongest baseline. On ImageNet-1K, Small and Base models reach 79.43% and 82.24% validation Top-1 accuracy. Four Long Range Arena tasks, diagnostics of trained operators, and CUDA timings further evaluate sequence accuracy, operator stability, and computational cost.

1 Introduction

Visual recognition and offline sequence understanding require representations that combine evidence from across an input. Pairwise attention provides this global interaction, but its quadratic arithmetic becomes costly as sequences grow. Compact memories offer a different computational route: encode the context in a bounded state and use that state to inform every output position. The challenge is to form a memory that captures the associations needed to interpret the current sample.

Test-time training (TTT) makes this adaptation explicit by treating contextual associations as training data for an internal model [31]. The outer network learns how to construct and read that model, while an inner computation adapts it to the current input. ViT³ brings this perspective to visual sequence modeling and studies how the inner learning procedure should be designed [12]. In its visual experiments, a full-batch inner update outperforms sequential mini-batch updates; the authors relate this finding to the causal dependencies and overwriting introduced by sequential updates. For tasks with complete input access, this finding supports adapting a shared memory using all positions together.

Computing this shared state introduces a second problem. In an error-correcting memory, each association’s residual depends on the current state, and these residuals in turn change that state. A full-context update evaluates this feedback across all positions, but a finite update budget still makes the result depend on the chosen step sizes and number of steps. ViT³ illustrates the practical tradeoff: additional full-batch inner updates can improve accuracy, while reducing throughput and, in some tested configurations, destabilizing training [12]. We therefore ask whether full-context memory correction can be formulated as a compact equilibrium problem with a direct solution and a controlled readout.

DeltaNet and MesaNet suggest how to formulate such a problem. DeltaNet updates memory using the residual between an association’s prediction and target, and efficient chunkwise algorithms enable parallel training [30, 41]. MesaNet instead uses conjugate gradients to solve a contextual regression problem at causal positions [35]. We draw on DeltaNet’s residual correction and MesaNet’s use of a solve to construct a shared full-context memory. The difficulty is to preserve this error-correcting mechanism while making

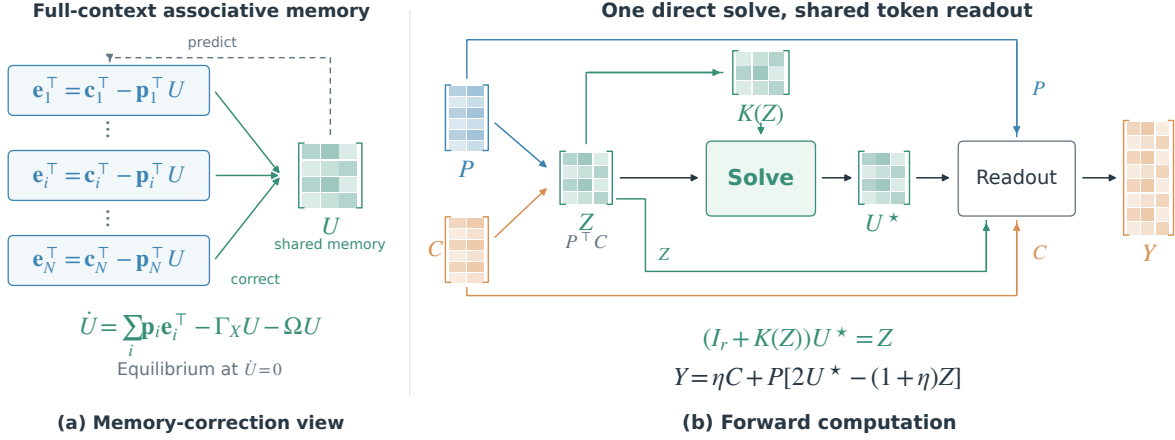


Figure 1: **Full-context associative memory in LSSO.** (a) Each association produces a residual under the same memory U . The weighted residuals, regularization, and skew coupling define the memory-correction field, with $\Gamma_X = \mathbf{I}_r + LL^T - P^T P$. (b) LSSO evaluates its equilibrium directly: P and C form the statistic Z , which generates $K(Z)$ and supplies the right-hand side of the solve. The readout combines U^* with the shared frame, content, and statistic. Panel (a) is an equivalent dynamics interpretation, not an unrolled computation. Blue denotes association coordinates, orange denotes content and output, and green denotes memory quantities.

full-context computation efficient. LION’s Appendix B.7 gives bidirectional recurrent and chunkwise DeltaNet formulations, but identifies a bottleneck in the full-attention expression it considers: a matrix inverse whose cost grows cubically with sequence length [1]. Thus, even though this expression exposes parallel computation, its $O(N^3)$ inversion can dominate the cost of processing long sequences and negate the intended training advantage. LION consequently avoids that full-attention route for DeltaNet training. The challenge is therefore to retain global association-error feedback without solving a system whose dimension grows with the sequence.

We introduce LSSO, which constructs a shared memory equilibrium from the complete input and evaluates it by one direct solve per sample and head. A soft association frame P gathers content C into a cross-moment $Z = P^T C$, which generates a compact system $K(Z)$. The memory satisfies

$$(\mathbf{I}_r + K(Z))U^* = Z. \quad (1)$$

All valid tokens jointly determine this system, but its dimension is the compact rank r , independent of sequence length N . The factorization and solve cost $O(r^3 + r^2 d)$ per head, and gathering and reading the context cost $O(Nr^2 + Nrd)$. This places the expensive solve in memory space while allowing all positions to participate, giving linear total cost in N at fixed rank. Strict accretivity makes the equilibrium unique and bounds the inverse used in the state and adjoint solves. For fixed sample-conditioned quantities, the system admits an exact decomposition into regularized association-error correction and skew-symmetric coupling. The forward pass computes this equilibrium without unrolling updates. A readout through the same association frame then gives each token access to the shared state. We choose the readout so that its spectral norm is strictly below one for each input and is exactly computable from compact matrices. This bound applies with the generated coefficients fixed; the full network Jacobian also includes derivatives of those coefficients. The implementation uses token-rank products and the small solve, giving linear sequence scaling at fixed rank without forming a dense token matrix.

The paper makes three contributions:

1. We construct a full-context memory layer with a directly solved equilibrium and derive its connection to regularized association-error correction.
2. We prove bounds on the state and adjoint solves, characterize the compact contraction family, and derive the exact spectral norm of the induced token mixer.

3. We evaluate the layer on ImageNet classification, genomic tasks, and Long Range Arena, measure operator margins in trained models, and implement an equivalent CUDA forward and backward for timing comparisons.

2 Related Work

Test-time training and full-context visual adaptation. TTT layers treat hidden states as internal models adapted on the current context [31]. ViT³ studies the inner objectives, update settings, and model architectures needed for visual TTT [12]. Its full-batch updates provide a noncausal form of adaptation. ZipMap likewise forms a shared scene state from an image collection [15]. LSSO defines a different internal system whose equilibrium admits a compact direct solve. Work on KV-binding TTT also shows why the quality of an inner solution must be distinguished from its usefulness for the outer task [22].

Error-correcting memory and solved adaptation. Fast-weight interpretations connect linear attention to associative memory; DeltaNet replaces purely additive storage with prediction-error correction [30, 41]. Test-time regression provides a common language for contextual memory objectives [36], and Longhorn derives state-space updates from online learning [21]. MesaNet studies contextual regression solved through conjugate gradients with a chunkwise implementation [35]. LSSO uses these ideas to construct an accretive system from the complete context. Section 3.5 derives the corresponding association-error dynamics, including the role of the skew term.

Bidirectional sequence computation. LION gives equivalent full, recurrent, and chunkwise views of bidirectional linear attention [1]. Its Appendix B.7 separates bidirectional DeltaNet formulations from the matrix-inverse cost of the full-attention expression it analyzes. LSSO constructs a different global system with a solve dimension set by compact rank. A related architectural route extends selective SSMS such as Mamba [9, 7]: Hydra uses quasiseparable matrix mixers to construct a principled bidirectional extension [13]. LSSO’s construction starts from a sample-generated memory equilibrium and its readout. Matched comparisons with these recent architectures remain future work.

Compact states and global interaction. Efficient attention uses projections, kernels, landmarks, or inducing states to avoid a dense token map [38, 17, 5, 28, 40]. Set Transformer, Luna, ABC, and Perceiver mediate global interactions through a bounded auxiliary representation [19, 24, 27, 14]. XCiT uses token-aggregated cross-covariance to generate a shared channel mixer [2]. In LSSO, a compact cross-moment generates the system to be solved, and the same association frame gathers the context and reads out the solution.

Monotone equilibria and contractive parameterization. Deep equilibrium models differentiate through implicit states [3], and monotone operator equilibrium networks impose structure that makes those states well posed [39]. Cayley relations between accretive operators and contractions are classical [4]; direct parameterizations also provide complete charts for static Lipschitz-bounded network weights [37]. LSSO uses these tools for a sample-generated memory equilibrium and its structured token-space readout. Its norm analysis fixes the generated coefficients, whereas global Lipschitz analyses of self-attention must also account for their input dependence [18].

Hardware-efficient execution. FlashAttention and hardware-aware SSM algorithms demonstrate the importance of memory traffic and execution structure alongside arithmetic complexity [6, 9]. DeltaNet’s chunkwise algorithm also makes error-correcting memory practical through hardware-efficient training [41]. LSSO lowers its compact equilibrium algebra and analytic backward to tiled CUDA kernels. At fixed rank its sequence-dependent arithmetic is linear, while exact softmax attention retains quadratic pairwise arithmetic even without materializing its score matrix. Section 5.5 reports complete-mixer measurements; Appendix E distinguishes the current implementation from the historical timing protocol.

3 LSSO: Full-Context Memory Adaptation

LSSO first gathers the input into association coordinates, then constructs and solves a compact memory system, and finally reads the state back to tokens. The memory-correction interpretation follows from this construction.

We describe one head and omit the batch index. Let $X \in \mathbb{R}^{N \times D}$ be the incoming token matrix, d the head width, and r the compact rank. A single input projection produces relation and content coordinates

$$[A, C] = XW_{bc} + b_{bc}, \quad A \in \mathbb{R}^{N \times r}, \quad C \in \mathbb{R}^{N \times d}. \quad (2)$$

Heads are concatenated and passed through an ordinary output projection after mixing. The construction below is repeated independently for each head, with one compact memory state shared by all valid tokens in that head. The state is recomputed from the current sample; learned outer parameters remain fixed during each forward evaluation.

3.1 Overview

For normalized, position-transformed relation coordinates, LSSO builds a soft frame P and evaluates

$$Z = P^\top C, \quad U^* = (\mathbf{I}_r + K(Z))^{-1}Z, \quad Y = \eta C + P[2U^* - (1 + \eta)Z]. \quad (3)$$

Here P is the association frame, K is generated from Z , and η is a learned scalar. We define each below, starting with the token coordinates used to construct P .

3.2 Relation features and masking

For a boolean validity mask $m \in \{0, 1\}^N$, invalid rows of both A and C are set to zero before any compact statistic is formed. Let

$$n = \sum_{i=1}^N m_i. \quad (4)$$

For a nonempty sequence, relation coordinates are normalized as $A \leftarrow A/\sqrt{n}$. An entirely masked sequence is assigned the zero output by definition. Masking before normalization makes the result invariant to appended padding.

The operator consumes relation and content features without an internal position-dependent transform. Absolute or two-dimensional learned position embeddings belong to the surrounding encoder.

3.3 Association frame

The QR soft frame maps unconstrained relation coordinates to a strict column contraction. Form the reduced QR factorization

$$\begin{bmatrix} P \\ E \end{bmatrix} R_{\text{qr}} = \begin{bmatrix} A \\ \mathbf{I}_r \end{bmatrix}, \quad \begin{bmatrix} P \\ E \end{bmatrix}^\top \begin{bmatrix} P \\ E \end{bmatrix} = \mathbf{I}_r, \quad (5)$$

where $P \in \mathbb{R}^{N \times r}$ is the token-space frame. A common positive rescaling, which leaves the exact Q factor unchanged, controls the numerical range. The identity augmentation means that the factorization is defined even when $N < r$ or A is rank deficient.

Each row of P supplies an association coordinate, and C supplies its content target. A memory $U \in \mathbb{R}^{r \times d}$ predicts those targets as PU . The frame compresses the content into one shared cross-moment,

$$Z = P^\top C \in \mathbb{R}^{r \times d}. \quad (6)$$

Z aggregates all valid positions symmetrically in one noncausal pass without forming the token-level matrix in (23).

Theorem 3.1 (Soft-frame spectrum and coverage). *Let P be the upper block of the Q factor in (5). If $\sigma_i(A)$ are the singular values of the normalized relation matrix, then*

$$\sigma_i(P) = \frac{\sigma_i(A)}{\sqrt{1 + \sigma_i(A)^2}}. \quad (7)$$

Consequently,

$$P^\top P \prec \mathbf{I}_r, \quad \mathbf{I}_N - PP^\top \succ 0, \quad 1 - \|P\|_2^2 = \frac{1}{1 + \|A\|_2^2}. \quad (8)$$

Moreover, at the level of represented frames, the construction covers every $P \in \mathbb{R}^{N \times r}$ satisfying $P^\top P \prec \mathbf{I}_r$.

The lower QR block satisfies $E = R_{\text{qr}}^{-1}$, so $P^\top P + E^\top E = \mathbf{I}_r$ gives the strict frame margin. Appendix A derives the spectrum and proves coverage of the represented frame family.

The identity augmentation supplies the strict complement $\mathbf{I}_N - PP^\top$, which allows the compact contraction to extend to every token direction. If A_0 denotes the masked relation matrix before division by $\sqrt{n}$, the final term in (8) becomes $1/(1 + \|A_0\|_2^2/n)$. Thus bounded per-token relation energy prevents the frame margin from deteriorating merely because the sequence is longer.

3.4 Memory system

In the default DYNAMIC mode, Z generates compact raw coordinates

$$\Theta = \Theta_0 + \frac{ZW_{\text{drive}}}{\sqrt{n}}, \quad (9)$$

where $\Theta_0 \in \mathbb{R}^{r \times r}$ and $W_{\text{drive}} \in \mathbb{R}^{d \times r}$ are learned per-head parameters. The STATIC ablation uses $\Theta = \Theta_0$, while ZERO fixes the compact contraction to zero.

Definition 3.2 (Strict accretivity). *A real square matrix K is strictly accretive when*

$$\text{sym}(K) := \frac{K + K^\top}{2} \succ 0. \quad (10)$$

For DYNAMIC and STATIC, define

$$L(\Theta) = \text{tril}(\Theta, -1) + \text{Diag}(\text{softplus}(\text{diag}(\Theta) + c_1)), \quad (11)$$

$$\Omega(\Theta) = \text{triu}(\Theta, 1) - \text{triu}(\Theta, 1)^\top, \quad (12)$$

$$K(\Theta) = L(\Theta)L(\Theta)^\top + \Omega(\Theta), \quad (13)$$

where $c_1 = \text{softplus}^{-1}(1)$. Thus L is lower triangular with positive diagonal, while Ω is skew-symmetric. The compact equilibrium and the pre-output token coordinates are

$$U^* = (\mathbf{I}_r + K)^{-1}Z, \quad (14)$$

$$Y = \eta C + P[2U^* - (1 + \eta)Z], \quad \eta = \tanh(\eta_{\text{raw}}) \in (-1, 1). \quad (15)$$

The complement is learned independently for each head and is initialized to 0.9. In ZERO, $M = 0$, which gives

$$Y = \eta(C - PZ). \quad (16)$$

At $\Theta = 0$, $L = \mathbf{I}_r$, $\Omega = 0$, $K = \mathbf{I}_r$, and the compact contraction is $M = 0$. Both nonzero-core modes start at that point; DYNAMIC also starts with $W_{\text{drive}} = 0$.

3.5 Association-error correction

For DYNAMIC and STATIC, the constructed system has a precise error-correction interpretation. Fix the current sample's P , C , L , and Ω , and define the derived regularizer

$$\Gamma_X = \mathbf{I}_r + LL^\top - P^\top P \succ 0. \quad (17)$$

Positive definiteness follows from Theorem 3.1 and $LL^\top \succ 0$. The coefficients are fixed during the solve. In particular, Γ_X is derived from the frame and system parameters rather than learned separately.

Proposition 3.3 (Associative-memory correction identity). *For the fixed contextual quantities above, let*

$$\mathcal{J}_X(U) = \frac{1}{2} \|PU - C\|_F^2 + \frac{1}{2} \text{tr}(U^\top \Gamma_X U). \quad (18)$$

Then the memory-correction field satisfies

$$\dot{U} = \underbrace{P^\top(C - PU)}_{\text{association-error feedback}} - \Gamma_X U - \Omega U \quad (19)$$

$$= -\nabla_U \mathcal{J}_X(U) - \Omega U = Z - (\mathbf{I}_r + K)U. \quad (20)$$

Its unique equilibrium is the state U^ used by LSSO.*

Proof. Since Γ_X is symmetric, $\nabla_U \mathcal{J}_X(U) = (P^\top P + \Gamma_X)U - P^\top C = (\mathbf{I}_r + LL^\top)U - Z$. Adding the skew term gives $Z - (\mathbf{I}_r + LL^\top + \Omega)U = Z - (\mathbf{I}_r + K)U$. The symmetric part of $\mathbf{I}_r + K$ is positive definite, so its zero is unique. $\square$

Writing row i of P as $\mathbf{p}_i^\top$ and row i of C as $\mathbf{c}_i^\top$ makes the feedback explicit:

$$P^\top(C - PU) = \sum_{i:m_i=1} \mathbf{p}_i(\mathbf{c}_i^\top - \mathbf{p}_i^\top U). \quad (21)$$

Every valid association contributes its prediction error to the same memory. When $\Omega = 0$, this is gradient flow on the strictly convex regularized association objective. With nonzero Ω , it is a monotone correction system with skew coupling; U^* need not minimize $\mathcal{J}_X$ itself. Appendix A.3 proves global convergence to U^* for the continuous process, while Section 4 bounds the solve and its implicit adjoint.

The forward pass obtains this equilibrium by solving the $r \times r$ system. Backpropagation differentiates through the solve and through the generation of its coefficients. The identity explains the memory corrected by LSSO; it does not require executing the continuous dynamics.

3.6 Token readout

Define the reflected resolvent

$$M = 2(\mathbf{I}_r + K)^{-1} - \mathbf{I}_r = (\mathbf{I}_r - K)(\mathbf{I}_r + K)^{-1}. \quad (22)$$

Substituting $Z = P^\top C$ into (15) gives the exact token-mixing form

$$Y = TC, \quad T = \eta \mathbf{I}_N + P(M - \eta \mathbf{I}_r)P^\top. \quad (23)$$

3.7 Linear sequence complexity

For one head, frame formation costs $O(Nr^2)$, the two frame-content contractions cost $O(Nrd)$, and construction and solution of the compact problem cost $O(r^2d + r^3)$. The forward arithmetic and activation/workspace storage are therefore

$$\mathcal{C}_{\text{LSSO}} = O(Nr^2 + Nrd + r^2d + r^3), \quad (24)$$

$$\mathcal{M}_{\text{LSSO}} = O(N(r + d) + r^2 + rd). \quad (25)$$

The analytic backward reuses compact factors and performs the same classes of token-rank contractions, so its arithmetic and saved-state dependence on N is also linear. Multiplying the head-local terms by the number of heads does not change the sequence-length degree.

Unlike exact MHA, whose $O(N^2d)$ mixing cost persists under FlashAttention [6], LSSO has $O(Nr(r + d))$ token-dependent work and no $N \times N$ intermediate. Both retain $O(ND^2)$ feature projections.

4 Properties of the Memory and Readout

We analyze the adaptation state and its induced readout for one realized head in exact arithmetic. We first bound the solve and its derivatives, then characterize the compact operator family and the spectral norm of the token readout. Complete proofs are given in Appendix A, and mixed-precision execution is described in Section 5.5.

4.1 Well-posed adaptation and implicit differentiation

Theorem 4.1 (Stable solved state and implicit adjoint). *Define the per-sample monotonicity margin*

$$\mu = 1 + \sigma_{\min}(L)^2 > 1, \quad A_K = \mathbf{I}_r + K. \quad (26)$$

Then $U^\star = A_K^{-1}Z$ exists uniquely, and

$$\sigma_{\min}(A_K) \geq \mu, \quad \|U^\star\|_F \leq \mu^{-1} \|Z\|_F, \quad \|A_K^{-\top} G\|_F \leq \mu^{-1} \|G\|_F. \quad (27)$$

For a second generated problem $A'_K = \mathbf{I}_r + K'$ with margin μ' and solution $U^{\star'} = A'^{-1}_K Z'$, the finite contextual perturbation obeys

$$\|U^\star - U^{\star'}\|_F \leq \mu^{-1} \|Z - Z'\|_F + (\mu\mu')^{-1} \|K - K'\|_2 \|Z'\|_F. \quad (28)$$

Consequently, an infinitesimal perturbation obeys

$$\|dU^\star\|_F \leq \mu^{-1} \|dZ\|_F + \mu^{-2} \|dK\|_2 \|Z\|_F. \quad (29)$$

The same computable margin controls the forward state, the transposed solve used by implicit differentiation, and sensitivity to the generated compact problem. The realized-gain certificate below separately measures the contraction margin, which also depends on the skew coordinates.

4.2 Context dependence and expressivity

Proposition 4.2 (Length-normalized contextual drive). *For the dynamic core,*

$$\|Z\|_F \leq \|P\|_2 \|C\|_F, \quad \|\Theta - \Theta_0\|_F \leq \frac{\|C\|_F}{\sqrt{n}} \|W_{\text{drive}}\|_2. \quad (30)$$

The first inequality is strict relative to $\|C\|_F$ whenever $C \neq 0$. Accordingly, if $\|C\|_F^2/n \leq c^2$, the magnitude of the contextual drive is at most $c \|W_{\text{drive}}\|_2$, independently of sequence length.

A complete compact contraction chart. The Cayley connection between accretive coordinates and contractions is classical in monotone operator theory [4], and direct Cayley charts have parameterized static Lipschitz-bounded network weights [37]. We use this relation to characterize the compact maps represented by LSSO.

Theorem 4.3 (Accretive Cayley parameterization). *The map in (13) ranges over all real strictly accretive $r \times r$ matrices. For every such K , the reflected resolvent M in (22) is a strict spectral-norm contraction and obeys*

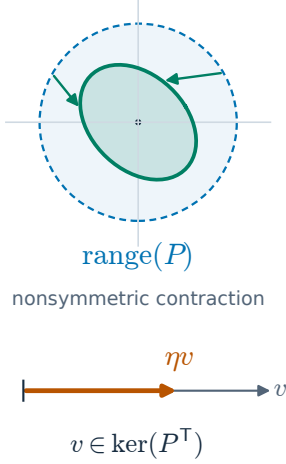
$$\mathbf{I}_r - M^\top M = 4(\mathbf{I}_r + K)^{-\top} L L^\top (\mathbf{I}_r + K)^{-1} \succ 0. \quad (31)$$

Conversely, the map $K \mapsto M$ is a bijection from real strictly accretive matrices to the open real matrix ball $\{M : \|M\|_2 < 1\}$, with inverse

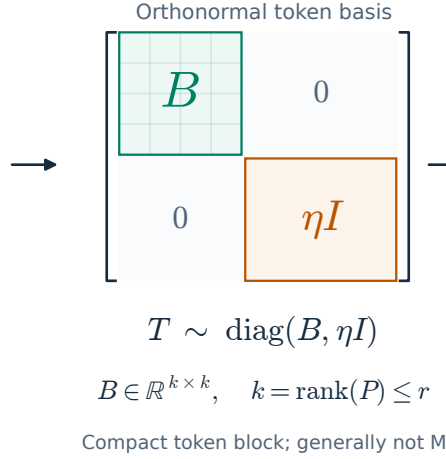
$$K = (\mathbf{I}_r - M)(\mathbf{I}_r + M)^{-1}. \quad (32)$$

Proof sketch. The construction has $\text{sym}(K) = L L^\top \succ 0$, because L has positive diagonal. Conversely, for any strictly accretive K , the Cholesky factor of $\text{sym}(K)$ and the skew remainder $K - \text{sym}(K)$ recover L and Ω , proving completeness.

a Token subspaces



b Compact representation



c Exact gain

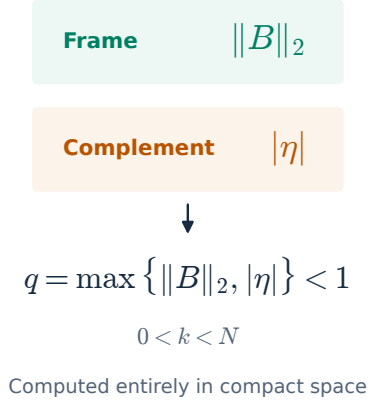


Figure 2: **From token subspaces to an exact gain.** (a) An illustrative nonsymmetric contraction in the frame subspace and scalar multiplication on its orthogonal complement; the scalar sketch uses positive η . (b) In an orthonormal token basis, the realized mixer separates into B and ηI . (c) For $0 < k < N$, its norm is the larger of the two block norms. Theorem 4.4 covers either sign of η and the empty and full-rank cases. The ellipse and matrix glyphs are schematic, not trained operator measurements.

Let $Q = \mathbf{I}_r + K$. Strict accretivity implies that Q is nonsingular. Using $M = (\mathbf{I}_r - K)Q^{-1}$ gives

$$\mathbf{I}_r - M^T M = Q^{-T} [Q^T Q - (\mathbf{I}_r - K)^T (\mathbf{I}_r - K)] Q^{-1} \quad (33)$$

$$= 4Q^{-T} \text{sym}(K)Q^{-1}, \quad (34)$$

which is (31). It is positive definite, so $\|M\|_2 < 1$. The inverse formula maps every strict contraction back to a strictly accretive matrix. $\square$

Because Ω is unrestricted, K and M may be nonsymmetric and nonnormal. The chart retains every strict contraction in rank space, and at token level, $T - T^T = P(M - M^T)P^T$, so this nonsymmetric structure survives the lift.

4.3 Exact gain of the global readout

The soft frame identifies the token subspace where contextual mixing acts. Figure 2 separates this compact action from the scalar complement. Because P is generally not orthonormal, the restricted token block B below differs from the compact Cayley core M .

Theorem 4.4 (Exact realized-gain certificate). *Fix a sample and head, and let P , M , and η be constructed as above. Write the positive eigenspace of $P^T P$ as*

$$P^T P = V_k S^2 V_k^T, \quad k = \text{rank}(P), \quad (35)$$

where $S \in \mathbb{R}^{k \times k}$ is positive diagonal, and define

$$B = \eta \mathbf{I}_k + S V_k^T (M - \eta \mathbf{I}_r) V_k S. \quad (36)$$

Then the singular values of T are exactly those of B , together with $N - k$ copies of $|\eta|$. Equivalently,

$$q(X) := \|T\|_2 = \begin{cases} |\eta|, & k = 0, \\ \sigma_{\max}(B), & k = N, \\ \max\{\sigma_{\max}(B), |\eta|\}, & 0 < k < N, \end{cases} \quad q(X) < 1. \quad (37)$$

Moreover, $\text{rank}(T - \eta \mathbf{I}_N) \leq r$ and $Tv = \eta v$ for every $v \in \ker(P^\top)$.

Proof sketch. Let $Q = PV_k S^{-1}$. Then $Q^\top Q = \mathbf{I}_k$, and substitution gives $T = QBQ^\top + \eta(\mathbf{I}_N - QQ^\top)$, so an orthogonal basis containing Q yields the stated singular spectrum. With $\Delta = (\mathbf{I}_N - PP^\top)^{1/2}$ and $F = [P \ \Delta]$,

$$T = F \text{Diag}(M, \eta \mathbf{I}_N) F^\top, \quad (38)$$

and $FF^\top = \mathbf{I}_N$; hence $\|T\|_2 < 1$. Finally, $T - \eta \mathbf{I}_N = P(M - \eta \mathbf{I}_r)P^\top$, which gives both the rank bound and $Tv = \eta v$ on $\ker(P^\top)$. $\square$

The exact certificate requires only an $r \times r$ Gram eigendecomposition and a singular-value calculation of size at most $r \times r$; no $N \times N$ kernel is needed. Because the forward uses the input-dependent $T(X)$ itself, Equation (37) gives the pointwise energy statement

$$\|T(X)C(X)\|_F \leq q(X) \|C(X)\|_F < \|C(X)\|_F \quad (39)$$

whenever $C(X) \neq 0$. The guarantee is pointwise in the input-conditioned operator. It does not bound the Jacobian of $X \mapsto T(X)C(X)$, which also contains $(dT)C$. Along any fixed realized sequence of mixer maps, the gains compose: $\|T_L \cdots T_1\|_2 \leq \prod_{\ell=1}^L q_\ell$.

Remark 4.5 (Why the scalar complement is part of the model). *Without the complement, a conventional form $\mathbf{I}_N + PAP^\top$ fixes the entire $\ker(P^\top)$ subspace to gain one. Equation (38) instead assigns that unmodeled subspace the learned per-head scalar η , while retaining the same rank- r correction and the strict contraction certificate. Thus T is a rank- r correction to an isotropic complement.*

Appendix A proves padding isolation and noncausal permutation equivariance.

5 Experiments

We evaluate LSSO in image classifiers and bidirectional sequence encoders. ImageNet training follows the DeiT III protocol. The accuracy and timing tables retain their original source revisions. Those runs included a position-dependent relation-feature transform that is absent from the current operator. They therefore describe the archived models, not new measurements of the current implementation. The archived genomic and LRA experiments use the DYNAMIC core, rank 32, FP16 automatic mixed precision, and the CUDA implementation. For these sequence experiments, checkpoints are selected only by validation accuracy, with validation loss breaking an exact tie; the held-out test split is evaluated once after selection. Reported uncertainties are sample standard deviations over seeds 0, 1, and 2. Across the genomic and LRA configurations, LSSO requires 1.3%–32.0% of exact MHA’s sequence-dependent mixer-core MACs (Appendix D).

5.1 ImageNet classification

Experimental settings. We evaluate image classification on ImageNet-1K, which contains 1.28M training images and 50K validation images from 1,000 classes [29]. We replace the attention layers in DeiT III with LSSO and follow the DeiT III training protocol [34]. Both Small and Base use 12 blocks, 16×16 patches, and a class-token readout. Their widths, head counts, and compact ranks are (384, 6, 32) and (768, 12, 48), respectively. All LSSO layers use the DYNAMIC core.

We train for 800 epochs with fused LAMB and a total batch size of 2048, using 3-Augment, Mixup, CutMix, and binary cross-entropy. Small is trained at 224×224 ; Base is trained at 192×192 and then fine-tuned for 20 epochs at 224×224 with AdamW and a batch size of 512. We report validation Top-1 accuracy at 224×224 . Checkpoint selection and full training settings are provided in Appendix B.

Table 1: ImageNet-1K classification at 224×224 . All models use ImageNet-1K training data only. Baseline results are quoted from the cited papers or official releases under their respective training protocols; LSSO follows DeiT III. Training schedules, augmentation, and other settings are not unified across methods.

Model	Mixing mechanism	Params (M)	Top-1 (%)
<i>Established classification baselines</i>			
DeiT-S [33]	Self-attention	22	79.8
DeiT-B [33]	Self-attention	86	81.8
DeiT III-S [34]	Self-attention	22.0	81.4
DeiT III-B [34]	Self-attention	86.6	83.8
ConvNeXt-T [23]	Convolution	28	82.1
ConvNeXt-B [23]	Convolution	89	83.8
<i>Bidirectional sequence mixers and visual TTT</i>			
Vim-S [42]	Bidirectional SSM	26	80.3
Vim-B [42]	Bidirectional SSM	98	81.9
Vim-S [†] [42]	Bidirectional SSM	26	81.4
Vim-B [†] [42]	Bidirectional SSM	98	83.2
Hydra-ViT-B [13]	Bidirectional SSM	91	81.6
LION-D [‡] (Small) [1]	Bidirectional linear attention	22	79.9
LION-D [‡] (Base) [1]	Bidirectional linear attention	86	80.2
ViT ³ -S [12]	Test-time training	24	81.6
ViT ³ -B [12]	Test-time training	90	82.6
<i>Our models</i>			
LSSO-S	Associative memory	19.62	79.43
LSSO-B	Associative memory	78.50	82.24

†: Vim fine-tuned for 30 additional epochs at 224×224 with patch size 16 and extraction stride 8. ‡: LION with multiple scans.

Results. Table 1 compares LSSO with established classifiers and related bidirectional sequence models. LSSO-S achieves 79.43% Top-1 with 19.62M parameters, and LSSO-B achieves 82.24% with 78.50M. LSSO-B is 0.44 points above the quoted DeiT-B result and 0.36 points below ViT³-B, while DeiT III-B reaches 83.8%. These comparisons place LSSO among existing visual backbones, but differences in training protocols prevent attributing the accuracy differences to the mixer alone.

5.2 Genomic classification

GenomicBenchmarks collects binary DNA sequence-classification tasks with official train and test partitions [8]. We derive a deterministic stratified validation set only from each training partition. All mixers use the same $D = 128$, four-layer, four-head encoder, learned absolute position embeddings, MLP ratio 4, mean readout, optimizer, data order, and 40-epoch budget; only the token mixer changes. Alongside MHA, we test bidirectional ELU + 1 Linear Transformer [17], FAVOR+ Performer with fixed orthogonal random features [5], and the official bidirectional ReLU/cosine cosFormer formulation [28]. Numerically sensitive attention cores run in FP32 under automatic mixed precision. The effective batch is 128, using two-step accumulation for Mouse Enhancers Ensembl. The three linear baselines match MHA’s parameter count; LSSO is 4.1% smaller on the 200-position tasks.

Table 2: GenomicBenchmarks held-out test accuracy (%), mean $\pm$ sample standard deviation over three seeds under the matched encoder and training protocol. Bold marks the highest mean in each row.

Task	MHA	Linear	Performer	cosFormer	LSSO
Coding vs. intergenomic	80.34 $\pm$ 1.99	78.11 $\pm$ 0.10	78.17 $\pm$ 0.01	79.12 $\pm$ 0.05	83.23 $\pm$ 2.46
Human or worm	65.87 $\pm$ 0.35	65.68 $\pm$ 0.22	65.76 $\pm$ 0.08	66.07 $\pm$ 0.20	81.57 $\pm$ 1.18
Mouse enhancers Ensembl	71.07 $\pm$ 1.65	71.63 $\pm$ 0.48	71.35 $\pm$ 0.48	71.35 $\pm$ 0.48	70.25 $\pm$ 0.83
Human enhancers Cohn	66.83 $\pm$ 0.17	66.89 $\pm$ 0.07	67.16 $\pm$ 0.08	67.06 $\pm$ 0.23	67.24 $\pm$ 0.26
Human enhancers Ensembl	76.37 $\pm$ 0.20	75.16 $\pm$ 0.57	74.32 $\pm$ 0.23	77.34 $\pm$ 1.53	78.91 $\pm$ 0.05
Human regulatory	92.14 $\pm$ 0.05	91.79 $\pm$ 0.15	90.10 $\pm$ 0.34	92.65 $\pm$ 0.14	93.10 $\pm$ 0.02
Human non-TATA promoters	84.71 $\pm$ 0.15	80.44 $\pm$ 1.84	82.51 $\pm$ 1.07	84.42 $\pm$ 0.48	85.41 $\pm$ 0.17
Human OCR Ensembl	65.61 $\pm$ 0.54	65.12 $\pm$ 0.26	65.36 $\pm$ 0.27	66.56 $\pm$ 0.07	68.55 $\pm$ 0.31
Unweighted task macro	75.37	74.35	74.34	75.57	78.53

Table 2 shows that LSSO has the highest mean on seven of eight tasks. Its 78.53 macro average exceeds cosFormer, the strongest alternative at 75.57, by 2.96 points and MHA’s 75.37 by 3.16 points. The largest improvement is on Human or Worm, while Linear Transformer leads on Mouse Enhancers Ensembl. The small mean differences on tasks such as Human Enhancers Cohn should be read alongside the three-seed variability. The macro average weights tasks equally after averaging seeds within each task.

5.3 Long Range Arena

Long Range Arena (LRA) tests structured reasoning and long-context classification [32]. Following established reporting practice [10], Table 3 quotes results from the original benchmark and subsequent work and groups models by their primary mixing mechanism. General-purpose global mixers and efficient-attention approximations form the first block; structured recurrence and state-space models form the second. These published results use different training protocols; locality and positional design also affect performance [26].

We evaluate LSSO on four LRA tasks using six-layer encoders, with $D = 128$ for ListOps and Retrieval and $D = 256$ for Text and Pathfinder-32. ListOps, Text, and Retrieval use learned one-dimensional absolute position embeddings with mean pooling; Pathfinder-32 uses factorized row/column embeddings and mean-maximum pooling. Their maximum sequence lengths are 2048, 4096, 4000, and 1024, respectively.

Table 3: LRA test accuracy (%) quoted from the cited sources under their reported protocols. The four-task average is unweighted over the displayed tasks. For LSSO, task scores and the per-seed four-task average are reported as mean $\pm$ sample standard deviation across three seeds. Bold marks the highest reported mean within each mechanism block.

Model and source	ListOps	Text	Retrieval	Pathfinder	4-task Avg.
<i>General-purpose global mixers and efficient-attention approximations</i>					
Transformer [32]	36.37	64.27	57.46	71.40	57.38
Linear Transformer [32]	16.13	65.90	53.09	75.30	52.61
Performer [32]	18.01	65.40	53.82	77.05	53.57
Linformer [32]	35.70	53.94	52.27	76.34	54.56
Reformer [32]	37.27	56.10	53.40	68.50	53.82
Luna-256 [24]	37.98	65.78	79.56	78.55	65.47
FNet [20]	35.33	65.11	59.61	77.80	59.46
Nyströmformer [40]	37.15	65.52	79.56	70.94	63.29
LSSO, three seeds	37.60 $\pm$ 0.65	65.19 $\pm$ 0.46	82.88 $\pm$ 0.08	78.79 $\pm$ 0.34	66.11 $\pm$ 0.24
<i>Models centered on structured recurrence or state-space dynamics</i>					
S4, original [10]	58.35	76.02	87.09	86.05	76.88
S4-LegS, updated [11]	59.60 $\pm$ 0.07	86.82 $\pm$ 0.13	90.90 $\pm$ 0.15	94.20 $\pm$ 0.25	82.88
Mega [25]	63.14	90.43	91.25	96.01	85.21

Within the global-mixer block, LSSO gives the highest Retrieval, Pathfinder, and four-task average; it is within 0.38 points of the highest ListOps result and 0.71 points of the highest Text result. S4 and Mega obtain higher scores across all four tasks. The table places LSSO among published results; the matched genomic experiment provides the controlled comparison of mixers.

5.4 Operator margins in trained models

We next test whether trained operators remain comfortably within the bounds of Section 4. Across three six-layer LRA Text checkpoints and matched held-out token-stream probes from 1K to 8K, all 9,216 head-level observations are finite and satisfy their certified conditions (Table 4). The worst realized gain is 0.9087, leaving at least 0.0913 contraction slack, and the minimum monotonicity margin is 1.1667. Solved states use at most 93.3% of their certified bound, while deterministic adjoint probes use at most 88.1%. Appendix C reports the per-layer distributions and stress-test protocol.

Table 4: Worst observation in the trained-model certificate audit. Bound usage is the measured ratio normalized by its certified upper bound $1/\mu$.

Quantity	Certified condition	Worst observed
Realized gain	$q(X) < 1$	$\max q = 0.9087$
Contraction slack	$1 - q(X) > 0$	$\min(1 - q) = 0.0913$
Monotonicity margin	$\mu > 1$	$\min \mu = 1.1667$
Solved-state bound usage	$\mu \ U^*\ _F / \ Z\ _F \leq 1$	0.9329
Adjoint-probe bound usage	$\mu \ A_K^{-T}G\ _F / \ G\ _F \leq 1$	0.8814

The gain in these probes is dominated by the scalar complement $|\eta|$ (Appendix C). The nearly constant worst-case gain therefore reflects the learned complement; it does not measure how strongly the dynamic core contributes to predictions. The 8K probes are operator stress tests on assembled token streams, not an 8K classification or length-generalization result.

5.5 Runtime

The DYNAMIC, STATIC, and ZERO cores have a native, algebraically equivalent CUDA path for inference and first-order training. It avoids both the $N \times N$ token matrix and the compact Cayley matrix; compact factorization and tiled solves evaluate the equilibrium directly, while the analytic backward reuses saved factors. Mixed-precision outputs and gradients are validated against the FP64 reference. Appendix E gives the execution schedule, workspace, VJP, numerical envelope, and build coverage.

These timings use the implementation and software versions recorded in Appendix E, rather than the subsequently revised backend. Table 5 measures the complete mixer boundary, including input and output projections. At $N = 4096$, native LSSO reaches $1.79\times$ the Flash-SDPA MHA forward throughput and $2.30\times$ its training throughput. Across $N = 1024$ – 4096 , forward-backward latency grows by $3.88\times$ for LSSO and $7.92\times$ for MHA. At $N = 1024$, LSSO is slower in inference (2.539 versus 2.131 ms), so the throughput advantage depends on sequence length. The timings cover the mixer, including projections, rather than the complete encoder.

Table 5: Synchronized milliseconds per batch on an RTX 5070 Ti (lower is better), with $B = 64$, $D = 256$, $H = 8$, FP16 AMP, and $r = 32$ for LSSO (historical contract 6). Ratios above one favor native LSSO; forward-backward includes loss and gradient reset. Bold marks the lowest latency in each row.

N	PyTorch ref.	Native	MHA	Ref./native	MHA/native
<i>Inference forward</i>					
1024	110.216	2.539	2.131	$43.41\times$	$0.84\times$
2048	133.940	5.577	5.888	$24.02\times$	$1.06\times$
4096	169.549	10.135	18.173	$16.73\times$	$1.79\times$
<i>Forward-backward with gradient reset</i>					
1024	118.858	7.512	8.487	$15.82\times$	$1.13\times$
2048	153.955	15.260	22.940	$10.09\times$	$1.50\times$
4096	211.268	29.160	67.195	$7.25\times$	$2.30\times$

6 Limitations

LSSO compresses the associations in a sequence into a memory of fixed rank. For each sample, the frame selects at most r token directions for nontrivial mixing; the orthogonal complement receives the same scalar action η . Although the frame adapts to the input, a fixed rank may limit the number of independent relations that one head can represent in a complex context. Increasing r expands this capacity, but raises the frame cost as $O(Nr^2)$ and the factorization cost as $O(r^3)$. Linear scaling with sequence length therefore comes with a capacity–computation tradeoff.

The direct solve also introduces overhead that is not amortized on every workload. In our measurements, LSSO has lower inference throughput than Flash-SDPA attention at $N = 1024$, with an advantage emerging at longer lengths. The crossover depends on model dimensions, precision, and hardware, so linear asymptotic complexity alone does not imply lower latency. The current architecture uses a single global memory per head; combining it with explicit local modeling is a useful direction for inputs whose fine-scale structure and long-range relations place different demands on the representation.

7 Conclusion

LSSO forms a shared full-context memory by directly solving a compact, input-dependent system. Its equilibrium can be expressed as regularized association-error correction with skew coupling. The system structure bounds the state and adjoint solves, while the frame readout yields a token mixer with an exactly computable spectral norm below one. The layer has linear sequence complexity at fixed rank and admits an equivalent CUDA implementation. ImageNet experiments demonstrate visual classification with the solved memory, matched genomic experiments show gains for the complete layer, and LRA results and diagnostics examine its behavior on longer sequences. Determining which parts of the construction account for these gains remains an important next step.

A Proofs and Additional Properties

A.1 Soft-frame spectrum and coverage

Let $A = USV^\top$ be an SVD of the normalized relation matrix. From the positive-diagonal QR factorization in (5),

$$R_{\text{qr}}^\top R_{\text{qr}} = \mathbf{I}_r + A^\top A, \quad P = AR_{\text{qr}}^{-1}. \quad (40)$$

Hence

$$PP^\top = A(\mathbf{I}_r + A^\top A)^{-1}A^\top = U \text{Diag}\left(\frac{\sigma_i(A)^2}{1 + \sigma_i(A)^2}\right)U^\top. \quad (41)$$

This proves (7), including rank-deficient A and $N < r$. The Cholesky/QR coordinates may differ from the symmetric-square-root form $A(\mathbf{I}_r + A^\top A)^{-1/2}$ by a right orthogonal factor, while their singular spectra and left Gram matrices agree.

Take any $P \in \mathbb{R}^{N \times r}$ with $P^\top P \prec \mathbf{I}_r$ and write $S = \mathbf{I}_r - P^\top P \succ 0$. Let C be the upper-triangular Cholesky factor of S , so $S = C^\top C$, and set $R_{\text{qr}} = C^{-1}$ and $A = PR_{\text{qr}}$. Then

$$\begin{bmatrix} A \\ \mathbf{I}_r \end{bmatrix} = \begin{bmatrix} P \\ R_{\text{qr}}^{-1} \end{bmatrix} R_{\text{qr}}. \quad (42)$$

The left factor has orthonormal columns because $R_{\text{qr}}^{-\top} R_{\text{qr}}^{-1} = S$. It is therefore the unique positive-diagonal reduced QR factorization and produces the desired upper block P .

A.2 Converse Cayley map and solve bound

For any M with $\|M\|_2 < 1$, the matrix $\mathbf{I}_r + M$ is nonsingular. Define $K = (\mathbf{I}_r - M)(\mathbf{I}_r + M)^{-1}$. Direct substitution gives

$$K + K^\top = 2(\mathbf{I}_r + M)^{-\top}(\mathbf{I}_r - M^\top M)(\mathbf{I}_r + M)^{-1} \succ 0. \quad (43)$$

Thus K is strictly accretive, and the forward and inverse fractional-linear expressions are algebraic inverses wherever defined.

Let μ be defined in (26). For a nonzero v , the skew part of K contributes zero to the quadratic form, so

$$\langle (\mathbf{I}_r + K)v, v \rangle = \|v\|_2^2 + \|L^\top v\|_2^2 \geq \mu \|v\|_2^2. \quad (44)$$

Cauchy–Schwarz therefore gives $\|(\mathbf{I}_r + K)v\|_2 \geq \mu \|v\|_2$, proving $\sigma_{\min}(\mathbf{I}_r + K) \geq \mu$ and $\|(\mathbf{I}_r + K)^{-1}\|_2 \leq \mu^{-1}$.

A.3 Convergence of the memory-correction dynamics

Let $A_K = \mathbf{I}_r + K$. By Proposition 3.3, the continuous process below is exactly the regularized association-error correction field with skew coupling. We establish its convergence while holding the sample-conditioned coefficients fixed.

Theorem A.1 (Convergence to the memory equilibrium). *For fixed contextual quantities K and Z , consider the continuous dynamics*

$$\dot{U}(t) = Z - A_K U(t). \quad (45)$$

From every initial state $U(0)$, the dynamics converge to $U^ = A_K^{-1}Z$ and obey*

$$\|U(t) - U^*\|_F \leq e^{-\mu t} \|U(0) - U^*\|_F. \quad (46)$$

The rate lower bound is independent of the skew-symmetric part of K .

Proof sketch. Write $E(t) = U(t) - U^*$. Since $A_K U^* = Z$, $\dot{E} = -A_K E$. The skew part vanishes in the Frobenius inner product, so

$$\frac{d}{dt} \frac{1}{2} \|E\|_F^2 = -\langle E, A_K E \rangle_F = -\langle E, \text{sym}(A_K) E \rangle_F \leq -\mu \|E\|_F^2, \quad (47)$$

because $\text{sym}(A_K) = \mathbf{I}_r + LL^\top \succeq \mu \mathbf{I}_r$. Gronwall’s inequality gives (46). $\square$

The next subsection gives a separate residual-minimization interpretation and derives the gradient of the solved state.

A.4 Residual optimization, implicit derivative, and sensitivity

The solved state also uniquely minimizes

$$J(U) = \frac{1}{2} \|A_K U - Z\|_F^2, \quad A_K = \mathbf{I}_r + K, \quad (48)$$

whose Hessian is $A_K^\top A_K \succ 0$. Gradient descent with $0 < \alpha < 2/\sigma_{\max}(A_K)^2$ obeys

$$\|U_j - U^*\|_F \leq \rho^j \|U_0 - U^*\|_F, \quad \rho = \max_i |1 - \alpha \sigma_i(A_K)^2| < 1. \quad (49)$$

Hence the direct solve is both the infinite-time limit in Theorem A.1 and the infinite-step limit of every such residual-gradient run.

Differentiating $A_K U^* = Z$ gives

$$dU^* = A_K^{-1}(dZ - dA_K U^*). \quad (50)$$

If $G = \partial \mathcal{L} / \partial U^*$ and $V = A_K^{-\top} G$, trace cyclicity yields the exact implicit VJP

$$\frac{\partial \mathcal{L}}{\partial Z} = V, \quad \frac{\partial \mathcal{L}}{\partial A_K} = -V U^{*\top}. \quad (51)$$

For two generated problems, the resolvent identity gives

$$U^* - U^{\star'} = A_K^{-1}(Z - Z') + A_K^{-1}(K' - K)A_K'^{-1}Z', \quad (52)$$

which proves (28). Taking its differential gives (50) and the stated local bound. The margin μ certifies the solve and flow rate, while the realized-gain certificate separately captures the effect of skew coordinates. For example, with $L = \mathbf{I}_2$ and $\Omega = a \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$, $\mu = 2$ for every a , while the associated $\|M\|_2$ approaches one as $|a| \rightarrow \infty$. This is why the realized token gain is computed separately by (37).

A.5 Exact compact token spectrum

With the notation of Theorem 4.4, $Q = PV_k S^{-1}$ has orthonormal columns. Substituting $P = QSV_k^T$ into (23) gives

$$T = QBQ^T + \eta(\mathbf{I}_N - QQ^T). \quad (53)$$

If $Q_\perp$ completes Q to an orthogonal matrix, then for $k > 0$

$$\begin{bmatrix} Q & Q_\perp \end{bmatrix}^T T \begin{bmatrix} Q & Q_\perp \end{bmatrix} = \begin{bmatrix} B & 0 \\ 0 & \eta \mathbf{I}_{N-k} \end{bmatrix}. \quad (54)$$

This proves the exact singular-spectrum union and the nonzero-rank cases of (37). The case distinction matters when $N < r$ and P has full row rank: then $k = N$, there is no token-space complement, and $|\eta|$ contributes no complement singular value. Since B may be nonnormal, its largest singular value, not its spectral radius, is the certificate. If $k = 0$, then $P = 0$ and $T = \eta \mathbf{I}_N$, giving the first case directly.

A.6 Padding isolation and permutation equivariance

If an invalid row has $A_i = C_i = 0$, the upper QR block has $P_i = 0$, so Equation (15) gives $Y_i = 0$. Explicit output masking preserves this value after the feature projection.

Let Π permute token rows together with their validity mask. All rowwise projections, masking, and normalization commute with Π . Uniqueness of the positive-diagonal reduced QR factorization gives $P' = \Pi P$, while $Z' = P'^T C' = P^T C = Z$. The generated K , M , and η are unchanged, and Equation (23) yields

$$\text{LSSO}(\Pi X, \Pi m) = \Pi \text{LSSO}(X, m). \quad (55)$$

B ImageNet Training Details

The three runs were recorded with source revision `bec1f18750ca4ad364b9d2a5b2ac63359b5cf6c9`, PyTorch 2.11.0+cu128, CUDA 12.8, and timm 1.0.28 on one NVIDIA H100 80GB HBM3. Training follows the DeiT III protocol and uses the `timm/imagenet-1k-wds` ImageNet-1K release with 1,281,167 training images and all 50,000 validation images. Repeated augmentation and batching produce 1,280,000 training presentations per initial-training epoch; fine-tuning uses 1,281,024. No external pretraining data or distillation teacher is used.

All models have 12 blocks, patch size 16, MLP ratio four, LayerScale initialization 10^{-4} , and normalization epsilon 10^{-6} . Small uses $(D, H, r) = (384, 6, 32)$ and constant stochastic depth 0.05; Base uses $(768, 12, 48)$ and stochastic depth 0.2. The class token has no learned position embedding, while the patch tokens do. Increasing Base resolution from 192 to 224 changes the recorded parameter count from 78,460,792 to 78,500,728 through the patch-position table; Small has 19,619,632 parameters.

Table 6: Recorded ImageNet optimization settings. Fine-tuning initializes from the best Base initial-training checkpoint.

Setting	Small initial	Base initial	Base fine-tuning
Resolution	224	192	224
Epochs	800	800	20
Optimizer	Fused LAMB	Fused LAMB	AdamW
Learning rate	0.004	0.003	10^{-5}
Minimum learning rate	10^{-5}	10^{-5}	10^{-5}
Weight decay	0.05	0.05	0.1
Effective batch	2048	2048	512
Physical batch / accumulation	1024/2	512/4	512/1
Augmentation	3-Augment	3-Augment	RandAugment
Repeated augmentation	Yes	Yes	No
Loss	BCE	BCE	Smoothed CE
Label smoothing	0	0	0.1

Each run uses seed 0, FP16 AMP, five warmup epochs from learning rate 10^{-6} , Mixup $\alpha = 0.8$, and CutMix $\alpha = 1.0$. Initial training uses a cosine schedule and color jitter 0.3; fine-tuning uses `rand-m9-mstd0.5-inc1`. EMA is maintained with decay 0.99996. The historical training loop calls validation on the ordinary model, not the EMA copy, and selects the best checkpoint by validation Top-1. Evaluation uses a crop ratio of 1.0. Top-5 accuracy is reported at the checkpoint selected by Top-1 accuracy.

Table 7: ImageNet validation results at the selected checkpoints. Epoch indices are one-based within each phase. The Base fine-tuning phase follows 800 initial epochs; these are stages of one training run, not independent seeds.

Run	Resolution	Best epoch	Top-1 (%)	Top-5 (%)
Small initial	224	759	79.434	94.290
Base initial	192	760	81.646	95.122
Base fine-tuning	224	15	82.240	95.944

Reference results. Table 1 uses the non-distilled DeiT models and the ImageNet-1K-only DeiT III and ConvNeXt releases. Hydra’s 81.6% is the EMA result in its Table 5; the non-EMA result is 81.0%. Other baselines retain their source papers’ evaluation settings. Vim’s long-sequence variants keep 224×224 images and 16×16 patches but reduce the extraction stride from 16 to 8 for 30 additional fine-tuning epochs. Thus their token count and computation differ from the standard variants even at the same image resolution.

C Operator Diagnostic Protocol

We use the validation-selected seed-0, seed-1, and seed-2 LRA Text checkpoints. Each is a six-layer, eight-head, rank-32 LSSO encoder trained at maximum length 4096. Sixteen probes are formed by concatenating the held-out validation token stream and taking matched prefixes of length 1024, 2048, 4096, and 8192. Thus a probe at one length is a prefix of the same probe at every longer length. This produces $3 \times 16 \times 6 \times 8 \times 4 = 9,216$ head-level observations.

At each layer, diagnostics are evaluated in FP64 from the realized compact factors. The exact gain $q(X)$ is obtained from the compact certificate block in Equation (36); no $N \times N$ operator is formed. We record the monotonicity margin μ , the normalized solved-state ratio $\|U^*\|_F / \|Z\|_F$, and the normalized adjoint ratio

$$\frac{\|A_K^{-\top} G\|_F}{\|G\|_F}, \quad A_K = \mathbf{I} + K. \quad (56)$$

Here G is a deterministic unit-Frobenius Rademacher matrix, shared across lengths for each probe-layer pair, that provides a reproducible probe of the implicit adjoint. Both ratios are checked against their shared upper bound $1/\mu$; equivalently, $\mu \|U^*\|_F / \|Z\|_F \leq 1$ and $\mu \|A_K^{-\top} G\|_F / \|G\|_F \leq 1$ are required for every observation.

We treat 8K as an operator-level stress test. The learned absolute position table is repeated modulo its 4096-position training range. The archived relation-feature preprocessing uses positions over the full probe. This protocol tests the mixer beyond its training length without introducing a new learned-position extrapolation rule.

Figure 3 shows the corresponding depth and length distributions. The realized-gain curves coincide because the learned scalar complement $|\eta|$ dominates Equation (37) in these checkpoints, yielding a stable worst-case gain across probe lengths. The compact-core diagnostics remain similarly stable: the monotonicity margin is preserved, the solved-state ratio stays bounded at 8K, and the adjoint-probe ratio remains near 0.50 across depth.

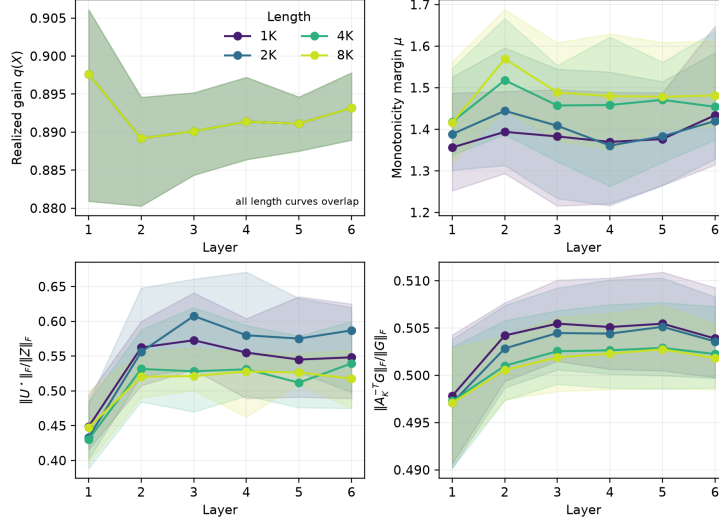


Figure 3: Per-layer certificate diagnostics from three trained LRA Text checkpoints. Curves show medians and shaded 10th–90th percentiles over held-out token-stream probes and heads. All four realized-gain curves overlap; the 8K points are operator-level stress measurements.

D Mixer-Core MAC Accounting

Table 8 reports the leading sequence-dependent matrix contractions per sample, accumulated over all mixer layers. For depth L , H heads, head width $d = D/H$, and rank $r = 32$, the accounting is

$$\text{MAC}_{\text{LSSO}} = 2LHNr(r + d), \quad \text{MAC}_{\text{MHA}} = 2LHN^2d. \quad (57)$$

The LSSO count comprises two token–rank frame passes and two frame–content contractions; the MHA count comprises $QK^\top$ and the attention–value product. Feature projections, elementwise maps, MLPs, task heads, and the N -independent compact factorizations and solves are excluded from both columns.

Table 8: Sequence-dependent mixer-core GMACs per sample at each task’s maximum length, accumulated over all encoder layers. The final column reports LSSO as a percentage of MHA.

Task	N	$D/L/H$	LSSO	MHA	LSSO/MHA (%)
<i>GenomicBenchmarks</i>					
Coding vs. intergenomic	200	128/4/4	0.0131	0.0410	32.0
Human or worm	200	128/4/4	0.0131	0.0410	32.0
Mouse enhancers Ensembl	4776	128/4/4	0.3130	23.3576	1.3
Human enhancers Cohn	500	128/4/4	0.0328	0.2560	12.8
Human enhancers Ensembl	573	128/4/4	0.0376	0.3362	11.2
Human regulatory	802	128/4/4	0.0526	0.6586	8.0
Human non-TATA promoters	251	128/4/4	0.0164	0.0645	25.5
Human OCR Ensembl	593	128/4/4	0.0389	0.3601	10.8
<i>Long Range Arena</i>					
ListOps	2048	128/6/8	0.3020	6.4425	4.7
Text	4096	256/6/8	0.8053	51.5396	1.6
Retrieval	4000	128/6/8	0.5898	24.5760	2.4
Pathfinder-32	1024	256/6/4	0.1510	3.2212	4.7

E CUDA Implementation Details

E.1 Current implementation contract

The current backend uses native ABI 11 and model checkpoint contract 13. It implements DYNAMIC, STATIC, and ZERO with the same masking and compact-state definition as Section 3. All three modes support inference and first-order training. The no-skew and no-complement variants require the reference backend; unsupported requests fail explicitly rather than selecting another backend.

The native path accepts FP16 or BF16 activations with FP32 parameters, ranks $r \in \{16, 32, 48, 64\}$, and positive head widths within allocation and launch limits. Boolean validity masks have shape $[B, N]$. Invalid rows are zeroed before compact statistics and again after output projection. Each sample uses $\sqrt{\max(n_{\text{valid}}, 1)}$ for normalization, including gapped masks and entirely masked samples. The mixer has no position-coordinate input. External position embeddings are applied by the encoder.

E.2 Base and correction evaluation

The reference evaluates the token output as

$$Y = \eta(C - PZ) + P\Delta, \quad \Delta = (\mathbf{I}_r + K)^{-1}(\mathbf{I}_r - K)Z. \quad (58)$$

The identity $(\mathbf{I}_r + K)^{-1}(\mathbf{I}_r - K) = 2(\mathbf{I}_r + K)^{-1} - \mathbf{I}_r$ establishes equality with the equilibrium readout. For STATIC, the reference computes this correction map once per head and applies it across the batch. ZERO sets $\Delta = 0$ and owns no compact-core parameters.

The native implementation retains the equivalent fused readout $Y = \eta C + P[2U - (1 + \eta)Z]$. It constructs the soft frame using a tiled Gram–Cholesky schedule algebraically equivalent to the positive-diagonal QR form, then computes $Z = P^T C$. DYNAMIC constructs and factors an input-conditioned system per sample and head. STATIC factors one system per head. For sufficiently many shared right-hand sides, it computes $S = (\mathbf{I}_r + K)^{-1}$ by solving against the identity once per head and applies $U = SZ$ across samples; smaller batches retain shared LU factors. The same stored map supplies the transposed action in backward. ZERO uses a literal zero correction and stores no core factors or equilibrium state. All modes share the coefficient $\Delta - \eta Z$ in the token readout, with $\Delta = 2U - Z$ for learned cores. No factors are cached across forward calls.

The public CUDA dispatcher uses an equivalent schedule that avoids materializing P . With $R^T R = \mathbf{I}_r + A^T A$ and $P = AR^{-1}$, it computes

$$Z = R^{-T} A^T C, \quad V = R^{-1}(\Delta - \eta Z), \quad Y = \eta C + AV. \quad (59)$$

The three core modes share token statistics, readout and input derivatives; only the compact correction differs. The explicit-frame native implementation is retained as a comparison baseline, not an automatic fallback. For $N \leq r$, the base term is evaluated as $\eta(\mathbf{I}_N + AA^T)^{-1}C$ to avoid subtracting nearly equal values. This system is at most $r \times r$, and neither schedule stores an attention matrix.

E.3 Backward and numerical boundaries

The analytic backward reuses saved factors for transposed solves and propagates gradients through the compact state, generator and soft frame. For STATIC, matrix adjoints are accumulated across samples before applying the shared parameterization Jacobian. The token-level frame and relation derivatives are fused to avoid storing a separate token-sized frame adjoint. The implementation supports first-order gradients; it does not differentiate through unrolled optimization steps.

Ordinary projection and eligible compact products use BF16 operands with FP32 accumulation. Packed native activations and their gradients are BF16. Frame storage, factorization, the accretive factor Gram, solve state and parameter gradients remain FP32. In the no-frame schedule, transformed compact coefficients are stored in FP32 and decomposed into a leading BF16 word and a BF16 residual word for token products, avoiding a single BF16 rounding of the combined coefficient. Its rank-space compact VJP is analytic, reusing cuSOLVERDx LU factors for transposed solves; the smaller token-system branch retains first-order autograd. The reference retains FP64 evaluation for output and gradient oracles. All three core modes are checked against this reference, including nonzero core parameters, gapped padding and fully masked samples. Kernel

and integration tests define the numerical acceptance criteria; implementation checks do not establish new task accuracy.

The current source targets PyTorch 2.14.0+cu132, CUDA 13.2 and MathDx 26.06.1. The updated build and oracle tests have been validated on SM80. The timing table below uses the earlier software and operator revision stated in its protocol; it is not a measurement of this updated backend.

E.4 Wall-clock protocol

Table 5 was measured under WSL2 on one NVIDIA GeForce RTX 5070 Ti (16 GB, SM120), driver 610.88, PyTorch 2.11.0+cu128, CUDA 12.8, and native contract version 6. Training measurements use training mode, while inference measurements use evaluation mode. Each process pregenerates a contiguous input tensor and enables TF32. LSSO uses the default DYNAMIC core with its archived relation-feature preprocessing, consecutive positions, and no padding mask. The comparison is PyTorch `nn.MultiheadAttention` with bias, zero dropout, batch-first layout, and `need_weights=False`, using PyTorch’s Flash SDPA forward and backward kernels.

The timed boundary includes both operators’ input and output projections, but excludes residual connections, MLPs, data loading, and optimizer updates. The forward–backward measurement differentiates a squared-mean scalar loss and includes setting the input gradient to `None` every iteration and model parameter gradients to `None` after each iteration. No CUDA Graph or `torch.compile` path is enabled. Every configuration runs 50 untimed warmup iterations, followed by seven repetitions of 100 iterations. Each repetition is bracketed by `torch.cuda.synchronize()`, and the reported number is the median synchronized host wall-clock time per iteration. Batch, width, head count, rank, precision, and statistical protocol are fixed across all sequence lengths and implementations. The benchmark entry point is `benchmarks/benchmark_ball.py`.

F Figure Sources and Production

The conceptual diagrams are supplied as vector PDFs. Their matrix textures and geometric illustrations are schematic and encode no experimental results. Layout and export review used the Scientific Visualization skill from Scientific Agent Skills [16]; the drawing code and operator content were written for this manuscript. The tensor-flow diagrams in ViT³ [12] and the semantic color grouping in LION [1] informed the visual organization; no source artwork was reused. The revised overview also draws on the state/update/readout organization of TTT [31] and the separation of parameter-generation and state-computation paths in Mamba [9]. All elements of the revised overview were drawn as vector primitives for LSSO. The existing certificate plot reports the diagnostic protocol in Appendix C.

References

- [1] Arshia Afzal, Elias Abad Rocamora, Leyla Naz Candogan, Pol Puigdemont, Francesco Tonin, Yongtao Wu, Mahsa Shoaran, and Volkan Cevher. Linear attention for efficient bidirectional sequence modeling. In *Advances in Neural Information Processing Systems*, volume 38, 2025.
- [2] Alaaeldin Ali, Hugo Touvron, Mathilde Caron, Piotr Bojanowski, Matthijs Douze, Armand Joulin, Ivan Laptev, Natalia Neverova, Gabriel Synnaeve, Jakob Verbeek, and Hervé Jégou. XCiT: Cross-covariance image transformers. In *Advances in Neural Information Processing Systems*, volume 34, pages 20014–20027, 2021.
- [3] Shaojie Bai, J. Zico Kolter, and Vladlen Koltun. Deep equilibrium models. In *Advances in Neural Information Processing Systems*, volume 32, 2019.
- [4] Heinz H. Bauschke and Patrick L. Combettes. *Convex Analysis and Monotone Operator Theory in Hilbert Spaces*. CMS Books in Mathematics. Springer, Cham, 2 edition, 2017.
- [5] Krzysztof Choromanski, Valerii Likhoshesterov, David Dohan, Xingyou Song, Andreea Gane, Tamas Sarlos, Peter Hawkins, Jared Davis, Afroz Mohiuddin, Lukasz Kaiser, David Belanger, Lucy Colwell, and Adrian Weller. Rethinking attention with performers. In *International Conference on Learning Representations*, 2021.
- [6] Tri Dao, Daniel Y. Fu, Stefano Ermon, Atri Rudra, and Christopher Ré. FlashAttention: Fast and memory-efficient exact attention with IO-awareness. In *Advances in Neural Information Processing Systems*, volume 35, pages 16344–16359, 2022.
- [7] Tri Dao and Albert Gu. Transformers are SSMS: Generalized models and efficient algorithms through structured state space duality. *arXiv preprint arXiv:2405.21060*, 2024.

- [8] Katarína Grešová, Vlastimil Martinek, David Čechák, Petr Šimeček, and Panagiotis Alexiou. Genomic benchmarks: A collection of datasets for genomic sequence classification. *BMC Genomic Data*, 24(1):25, 2023.
- [9] Albert Gu and Tri Dao. Mamba: Linear-time sequence modeling with selective state spaces. In *Conference on Language Modeling*, 2024.
- [10] Albert Gu, Karan Goel, and Christopher Ré. Efficiently modeling long sequences with structured state spaces. In *International Conference on Learning Representations*, 2022.
- [11] Albert Gu, Isys Johnson, Aman Timalsina, Atri Rudra, and Christopher Ré. How to train your HiPPO: State space models with generalized orthogonal basis projections. In *International Conference on Learning Representations*, 2023.
- [12] Dongchen Han, Yining Li, Tianyu Li, Zixuan Cao, Ziming Wang, Jun Song, Yu Cheng, Bo Zheng, and Gao Huang. ViT³: Unlocking test-time training in vision. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 51–61, 2026.
- [13] Sukjun Hwang, Aakash Lahoti, Ratish Puduppully, Tri Dao, and Albert Gu. Hydra: Bidirectional state space models through generalized matrix mixers. In *Advances in Neural Information Processing Systems*, volume 37, 2024.
- [14] Andrew Jaegle, Sebastian Borgeaud, Jean-Baptiste Alayrac, Carl Doersch, Catalin Ionescu, David Ding, Skanda Koppula, Daniel Zoran, Andrew Brock, Evan Shelhamer, et al. Perceiver IO: A general architecture for structured inputs and outputs. In *International Conference on Machine Learning*, 2021.
- [15] Haian Jin, Rundi Wu, Tianyuan Zhang, Ruiqi Gao, Jonathan T. Barron, Noah Snaveley, and Aleksander Holynski. ZipMap: Linear-time stateful 3D reconstruction via test-time training. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026.
- [16] Timothy Kassis, Vinayak Agarwal, Yuhuan He, Darshil Patel, and Aubrey M. Brueckner. Scientific agent skills: A library of procedural knowledge for research agents. *arXiv preprint arXiv:2609.00065*, 2026.
- [17] Angelos Katharopoulos, Apoorv Vyas, Nikolaos Pappas, and François Fleuret. Transformers are RNNs: Fast autoregressive transformers with linear attention. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, pages 5156–5165. PMLR, 2020.
- [18] Hyunjik Kim, George Papamakarios, and Andriy Mnih. The lipschitz constant of self-attention. In *International Conference on Machine Learning*, pages 5562–5571, 2021.
- [19] Juho Lee, Yoonho Lee, Jungtaek Kim, Adam R. Kosiorek, Seungjin Choi, and Yee Whye Teh. Set transformer: A framework for attention-based permutation-invariant neural networks. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 3744–3753. PMLR, 2019.
- [20] James Lee-Thorp, Joshua Ainslie, Ilya Eckstein, and Santiago Ontanon. FNet: Mixing tokens with fourier transforms. In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 4296–4313. Association for Computational Linguistics, 2022.
- [21] Bo Liu, Rui Wang, Lemeng Wu, Yihao Feng, Peter Stone, and Qiang Liu. Longhorn: State space models are amortized online learners. *arXiv preprint arXiv:2407.14207*, 2024.
- [22] Junchen Liu, Sven Elfle, Or Litany, Zan Gojcic, and Ruilong Li. Test-time training with KV binding is secretly linear attention. *arXiv preprint arXiv:2602.21204*, 2026.
- [23] Zhuang Liu, Hanzi Mao, Chao-Yuan Wu, Christoph Feichtenhofer, Trevor Darrell, and Saining Xie. A ConvNet for the 2020s. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022.
- [24] Xuezhe Ma, Xiang Kong, Sinong Wang, Chunting Zhou, Jonathan May, Hao Ma, and Luke Zettlemoyer. Luna: Linear unified nested attention. In *Advances in Neural Information Processing Systems*, volume 34, 2021.
- [25] Xuezhe Ma, Chunting Zhou, Xiang Kong, Junxian He, Liangke Gui, Graham Neubig, Jonathan May, and Luke Zettlemoyer. Mega: Moving average equipped gated attention. In *International Conference on Learning Representations*, 2023.
- [26] Pablo Miralles-González, Javier Huertas-Tato, Alejandro Martín, and David Camacho. On the locality bias and results in the long range arena. *arXiv preprint arXiv:2501.14850*, 2025.
- [27] Hao Peng, Jungo Kasai, Nikolaos Pappas, Dani Yogatama, Zhaofeng Wu, Lingpeng Kong, Roy Schwartz, and Noah A. Smith. ABC: Attention with bounded-memory control. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 7469–7483, Dublin, Ireland, 2022. Association for Computational Linguistics.

- [28] Zhen Qin, Weixuan Sun, Hui Deng, Dongxu Li, Yunshen Wei, Baohong Lv, Junjie Yan, Lingpeng Kong, and Yiran Zhong. cosFormer: Rethinking softmax in attention. In *International Conference on Learning Representations*, 2022.
- [29] Olga Russakovsky, Jia Deng, Hao Su, Jonathan Krause, Sanjeev Satheesh, Sean Ma, Zhiheng Huang, Andrej Karpathy, Aditya Khosla, Michael Bernstein, Alexander C. Berg, and Li Fei-Fei. ImageNet large scale visual recognition challenge. *International Journal of Computer Vision*, 115:211–252, 2015.
- [30] Imanol Schlag, Kazuki Irie, and Jürgen Schmidhuber. Linear transformers are secretly fast weight programmers. In *International Conference on Machine Learning*, 2021.
- [31] Yu Sun, Xinhao Li, Karan Dalal, Jiarui Xu, Arjun Vikram, Genghan Zhang, Yann Dubois, Xinlei Chen, Xiaolong Wang, Sanmi Koyejo, Tatsunori Hashimoto, and Carlos Guestrin. Learning to Learn at test time: RNNs with expressive hidden states. *arXiv preprint arXiv:2407.04620*, 2025.
- [32] Yi Tay, Mostafa Dehghani, Samira Abnar, Yikang Shen, Dara Bahri, Philip Pham, Jinfeng Rao, Liu Yang, Sebastian Ruder, and Donald Metzler. Long range arena: A benchmark for efficient transformers. In *International Conference on Learning Representations*, 2021.
- [33] Hugo Touvron, Matthieu Cord, Matthijs Douze, Francisco Massa, Alexandre Sablayrolles, and Hervé Jégou. Training data-efficient image transformers and distillation through attention. In *International Conference on Machine Learning*, pages 10347–10357, 2021.
- [34] Hugo Touvron, Matthieu Cord, and Hervé Jégou. DeiT III: Revenge of the ViT. *arXiv preprint arXiv:2204.07118*, 2022. ImageNet-1K model results: https://github.com/facebookresearch/deit/blob/main/README_revenge.md.
- [35] Johannes von Oswald, Nino Scherrer, Seijin Kobayashi, Luca Versari, Songlin Yang, Sarthak Mittal, Maximilian Schlegel, Kaitlin Maile, Yanick Schimpf, Oliver Sieberling, Alexander Meulemans, Rif A. Saurous, Guillaume Lajoie, Charlotte Frenkel, Razvan Pascanu, Blaise Agüera y Arcas, and João Sacramento. MesaNet: Sequence modeling by locally optimal test-time training. In *International Conference on Learning Representations*, 2026.
- [36] Ke Alexander Wang, Jiaxin Shi, and Emily B. Fox. Test-time regression: A unifying framework for designing sequence models with associative memory. *arXiv preprint arXiv:2501.12352*, 2025.
- [37] Ruigang Wang and Ian R. Manchester. Direct parameterization of lipschitz-bounded deep networks. *arXiv preprint arXiv:2301.11526*, 2023.
- [38] Sinong Wang, Belinda Z. Li, Madian Khabza, Han Fang, and Hao Ma. Linformer: Self-attention with linear complexity. *arXiv preprint arXiv:2006.04768*, 2020.
- [39] Ezra Winston and J. Zico Kolter. Monotone operator equilibrium networks. In *Advances in Neural Information Processing Systems*, 2020.
- [40] Yunyang Xiong, Zhanpeng Zeng, Rudrasis Chakraborty, Mingxing Tan, Glenn Fung, Yin Li, and Vikas Singh. Nyströmformer: A nyström-based algorithm for approximating self-attention. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 35, pages 14138–14148, 2021.
- [41] Songlin Yang, Bailin Wang, Yu Zhang, Yikang Shen, and Yoon Kim. Parallelizing linear transformers with the delta rule over sequence length. *arXiv preprint arXiv:2406.06484*, 2025.
- [42] Lianghui Zhu, Bencheng Liao, Qian Zhang, Xinlong Wang, Wenyu Liu, and Xinggang Wang. Vision mamba: Efficient visual representation learning with bidirectional state space model. In *International Conference on Machine Learning*, pages 62429–62442, 2024.